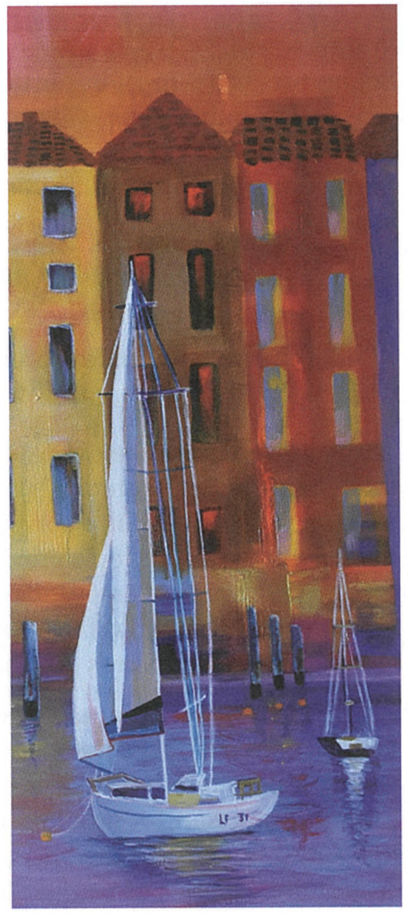
Discovering Economics

Contents

1 What is economics?



Learning Objectives

In this chapter you will learn about:

- *the meaning of economics*
- *scarcity and the economic problem*
- *the distinction between microeconomics and macroeconomics*
- *the different types of resources*
- *the concept of scarcity and tradeoffs*
- *the meaning of opportunity cost*
- *the use of economic models*
- *the distinction between positive and normative economics*
- *the production possibility frontier model*

The meaning of economics

Economics is the most important subject you will ever study! Economics is also the most useful subject you will ever study! Economics will provide you with important insights into how the world operates and will also help you in making decisions throughout your entire life. Economics is important because it permeates every aspect of life. Economic issues are everywhere. Consider, for example:

- global warming and climate change
- water restrictions
- university entrance scores
- the value of the $A

Why is there a shortage of hospital beds in a country as wealthy as Australia? Why is there a shortage of water in Australia? Why do you need an entrance score to gain a place at university? Will I get a job when I graduate? Why are some countries poor and others rich? Why do some people earn more income than others? Should there be a price on carbon emissions? It is impossible to escape economic issues and that is why everyone should have some basic economic literacy in order to understand and participate in today's economic world.

What is the subject matter of economics? Economics deals with two basic facts - first, people are faced with **limited resources** and second they have **unlimited wants**. This is referred to as the basic **economic problem**. The economic problem is on the one hand a problem of **scarcity** and on the other, a problem of **choice**. It is a problem of scarcity because there are simply not enough resources to satisfy an infinite number of wants. Individuals have basic wants of food, clothing, and housing. They also have other wants including transport, entertainment, and leisure. In economics we say that resources are scarce relative to wants. Scarcity is the most fundamental concept in economics! Almost everything is scarce - food, cars, water, iPhones, computers. Anything that has a price is scarce. Free goods are not relatively scarce and do not have a price. The air that you breathe is currently free but perhaps in the future it may have a price.

Global warming and climate change is a good example of the economic problem. The atmosphere has over time become a scarce commodity. Carbon dioxide emissions are continuously pumped into the atmosphere by industry, agriculture and households. Almost all production and consumption involves the use of fossil fuels - driving cars, food production, factory output and using electricity for lights and household appliances. The world relies on carbon based energy resources - coal, oil and natural gas. The use of these resources imposes a major environmental cost in terms of increased greenhouse gases in the atmosphere which raises global temperatures and changes world climate.

It is important to understand that the word scarcity in economics does not have the same meaning as the dictionary definition. For example, cars are plentiful in Australia (they are not physically scarce) - in 2019, over one million new cars were

sold in Australia. However, an economist would say that cars are relatively scarce because they are not free - everyone who wants a car may not necessarily be able to afford one. Cars are not given away for free to anybody who wants one! This applies to most goods and services. There are some government services which are provided free but the cost of these services are paid for out of tax revenue - public health and public education, roads and footpaths are examples. So even these goods are scarce.

It is important not to confuse the concept of scarcity with the word shortage. Almost everything is scarce - everything you buy that has a price is scarce. A shortage occurs when the supply of something is limited. Cars are a scarce economic good, but there is not a shortage of cars in Australia. Scarcity is also not the same thing as poverty. Poverty is a lack of income to meet basic economic needs. Rich and poor nations alike suffer from the economic problem. The richest person in the world may be able to satisfy all their basic needs but they will still suffer from a limited amount of time.

Economics is the study of how people allocate their limited resources to satisfy their unlimited wants. Economics is the study of the economic problem - it is a study of scarcity and a study of choice

The second aspect of the economic problem is the problem of choice. Given that scarcity exists for all people, consumers, producers and society must decide which wants they will satisfy. Should I spend my $20 going to the movies, purchasing songs from itunes, or buy a lotto ticket? Should the government spend more money on health, education or the environment? Making choices when confronted by scarcity involves a trade-off. Every society must make choices about what goods and services should be produced, how these goods and services will be produced and who will receive the goods and services. These are referred to as the three basic questions that any economic system must answer:

- what to produce?
- how to produce?
- for whom to produce?

One way to describe the subject matter of economics is that it is concerned with the production, distribution and consumption of goods and services. In other words, economics is concerned with the allocation of society's scarce resources and the distribution of income. Figure 1.1 lists several definitions of economics. Most of these definitions focus on the problem of scarcity.

Adam Smith (1723-1790) is regarded as one of the founders of economics, in recognition of his 1776 work entitled *"An Inquiry into the Nature and Causes of The Wealth of Nations"*. Smith extolled the virtues of the free market economy as being an efficient way to deal with the economic problem and highlighted the importance of specialisation. A simple definition of economics is that it is the study of how people and society deal with the basic problem of scarcity. In other words, *economics is the study of how people allocate their limited resources to satisfy their unlimited wants.* Economics is a **social science** because it uses the scientific method to investigate and analyse human behaviour. Economists develop and use economic models to formulate theories about how people make decisions. People have unlimited wants and the resources or factors of production to satisfy these wants are limited. Resources are therefore said to be scarce relative to wants - we simply do not have enough of them to satisfy the infinite number of wants. Read the definitions in figure 1.1 and identify the common elements.

Economic matters affect all people at all ages, and in all walks of life. As Alfred Marshall noted, economics is a study of the ordinary business of life - how to allocate time between work, leisure and study, and how to spend income. Another famous economist, John Maynard Keynes, regarded economics as more of a thinking process rather than as a body of knowledge. Economics provides a very useful framework for making decisions when confronted with scarcity.

What are resources?

Economists identify three general types of **resources**, or **factors of production**:

- natural resources
- human resources
- capital resources

Figure 1.1 Definitions of economics

Adam Smith	*"Economics is an enquiry into the nature and causes of the wealth of nations".*
Wikipedia	*"Economics is the social science that studies the production, distribution, and consumption of goods and services."*
Alfred Marshall	*"Economics is a study of mankind in the ordinary business of life .... it is on the one side a study of wealth; and on the other, the more important side, a part of the study of man".*
John Maynard Keynes	*"Economics is a method rather than a doctrine; an apparatus of the mind, a technique of thinking which helps its possessor to draw correct conclusions".*
Paul Samuelson	*"Economics is the study of how people and society choose to employ scarce productive resources, which could have alternative uses, to produce various commodities, and the distribution of these commodities among people and groups".*
Milton Friedman	*"Economics is the science of how a particular society solves its economic problems".*
Jacob Viner	*"Economics is what economists do".*
Joan Robinson	*"The purpose of studying economics, is not to devise a set of ready-made answers to economic problems, but to learn how to avoid being deceived by economists"*
International Encyclopedia of Social Sciences	*"Economics is the study of the allocation of scarce resources among unlimited and competing uses".*

Natural resources are the "gifts of nature" - resources such as air, water, minerals, energy resources such as coal and oil, and soil and vegetation. Basically any resource that is supplied from the natural environment. **Human resources** refers to the quantity and quality of the labour force. Human resources can be divided into **labour** (the physical or mental effort applied in the production of a good or service) and **enterprise** (the coordination and management of production by an entrepreneur). Enterprise also represents the ideas and skills which create products, incomes and profits.

Capital refers to the man-made resources which assists human resources in the production of goods and services. In economics, capital refers to physical capital and not financial capital. Physical capital is the 'tools of trade', the machinery and equipment that is required to produce goods and services. Examples of capital include the pen used by a journalist to write a story, an accountant's computer, a truck used to haul ore at a mine site and a crane that is used to help construct a building. An important type of capital is **social overhead capital** - this is the basic infrastructure of the economy, such as transport and communications, power and water supply, schools and hospitals. This essential infrastructure is usually supplied by the government.

Capital plays an important role in a modern economy. In economics, the creation of new capital goods is referred to as **investment**. Investment is not the purchase of shares or putting money into an interest bearing deposit. Investment is the creation of new machinery, new buildings, and new roads. Investment is seen as the engine of economic growth. Modern industrial economies are characterised by high rates of investment. Providing workers with more and better capital equipment helps to increase productivity and raise living standards.

Microeconomics and macroeconomics

The subject matter of economics is usually subdivided into microeconomics and macroeconomics. **Microeconomics** deals with the economic problem from an individual or 'micro' point of view. The prefix 'micro' refers to a small perspective. Microeconomics attempts to understand how consumers and producers make decisions Microeconomics studies how markets and prices work to allocate resources between all the competing industries in the economy. At the heart of microeconomics is the theory of demand and supply. Part One of this text (chapters 1-5) covers the most important aspects of microeconomics.

Macroeconomics deals with the economic problem from society's point of view. The prefix 'macro' refers to a large perspective. Macroeconomics is concerned with the performance of the whole economy. It focuses on total economic activity in terms of total production, total employment and the overall price level. The two most important parts of macroeconomics concern the theory of economic growth and business cycles. Economic growth is important because it determines a nation's standard of living. Economies also experience fluctuations in economic activity - the business cycle - which affect a country's rate of inflation and unemployment. The 'king' of economic indicators is GDP - gross domestic product. This measures the total value of final goods and services produced in an economy. The other two key macroeconomic indicators are the inflation rate and the unemployment rate. Part Two of this text (chapters 6 to 12) deals with the important aspects of macroeconomics.

The science of economics

Economics is a **social science**. It studies the behaviour of people and the choices they make in response to the economic problem. Economics is a decision making science - it is particularly concerned with analysing and explaining decisions concerning production, consumption and resource use. Economics differs from the physical sciences, such as chemistry and physics, in that it cannot conduct controlled experiments. It does, however, use a logical set of processes (the **scientific method**) to identify the principles that govern economic behaviour in society.

The scientific method begins with observation and hypothesis. An economist might want to analyse household petrol consumption over time. Some important variables affecting petrol consumption would be the price of petrol, consumers' income, and the price of other related goods, such as cars and public transport. The effect of each independent variable on petrol consumption must be analysed separately to determine its relative importance. For example, we could hypothesise that people purchase more petrol when its price falls - a negative relationship; or that people buy more petrol when the price of cars fall. Our hypotheses are valid only if other independent variables are kept constant. This is an example of the **ceteris paribus** assumption. This is a Latin term which means 'all other things constant'. We could conclude that the price of petrol and the quantity of petrol consumed is a negative relationship, as long as other variables affecting petrol consumption do not change.

Economic theory concerns explaining cause and effect between two or more variables. If variables A, B and C all affect variable Y, we must first investigate the effect of variable A by holding B and C constant; then we would analyse the effect of variable B by holding A and C constant and then variable C by holding A and B constant. We are using the ceteris paribus assumption to isolate the cause and effect of each separate variable.

Economists develop models to derive their theories about how the economy works. An **economic model** is a simplified representation of economic reality showing the relationship between certain economic variables. Using simplified models enables economists to determine cause and effect. The test of a 'good' model is its ability to predict behaviour. Economic theories focus on the most important variables that determine economic behaviour. This is referred to as abstraction. All unnecessary detail is removed so that the model represents a simplification of reality. A simple analogy of a model is a map. A road map includes only the essential information required to enable a traveller to get from one destination to another. As a traveller's model, the map simplifies the task of getting from point A to B. Figure 1.2 summarises how the scientific method is used to build models in economics. The usefulness of an economic model or theory lies in its ability to predict events accurately. If the model's

predictions are verified through observation and data collection, then the model is considered a good one. If not, then the model must be rejected and reformulated.

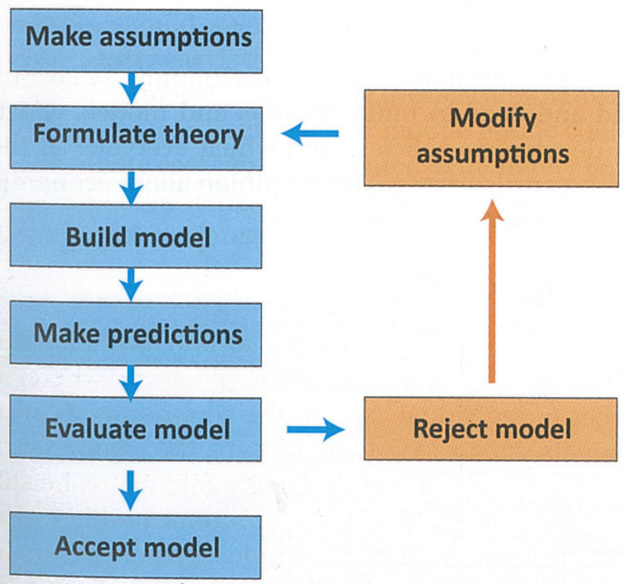
One of the important assumptions used by economists to explain human behaviour is that of **rational self-interest**. This means that economic decisions are based on a person following a logical process in order to compare the costs and benefits of decisions. A decision is considered rational as long as the benefit exceeds the cost. I will not purchase a good unless the benefit I get is greater than my cost. Price is a convenient way to measure and compare benefits and costs. If the price of a good rises, then for some people the higher cost will exceed their expected benefit and they will not consume the good. People will respond to incentives that affect the cost and benefits of a particular action. For example if the penalties for committing crime increase, then ceteris paribus, we would expect a decrease in crime. Economists believe that rational self-interest is a very good predictor of human behaviour. In fact, the model of demand and supply that is used to determine price in competitive markets is an excellent application of the cost/benefit approach.

Economists assume that people are rational and respond to incentives. A rational person is someone who compares the costs and benefits of a decision before acting. Rational people respond to incentives. If the price (cost) of a good falls, people will consume more.

Positive and normative economics

Economists not only develop economic theories and models to explain economic behaviour but they also formulate economic policy. Testing and developing economic theory is known as **positive economics**. Essentially, positive economics

Figure 1.2 Economic models



Assumptions are often used to simplify the real world. Theories are the result of observations of economic actions - they describe cause and effect relationships. Economic models are simplified descriptions of theories about the way the economy works. A good model provides accurate explanations of the relationships between different variables.

is concerned with 'what is' in the economy. Economics also deals with 'what should be' which involves personal opinion and value judgments. This is the domain of **normative economics.** Economic policy, while based on economic theory also involves normative economics. This is why economists will often disagree about economic policy. Positive economic statements can be tested objectively. What is the current unemployment rate? What would happen if tariffs are removed from imported cars? What effect will a sales tax on wine have on the wine industry? The answers to questions such as these are the subject of positive economics because they are testable using economic data.

Normative statements, on the other hand, are subjective statements which reflect opinions rather than facts, so they cannot be tested objectively. Normative statements often state 'what should be'. A normative statement involves a **value judgment** - an opinion that one situation is preferable to another. Opinions might appear logical, but they include a personal viewpoint which cannot be readily tested. Is the unemployment rate too high? Should the government protect domestic industry from foreign competition? Should the government tax carbon emissions? Each of these is a normative question – they invite an opinion. The two statements below highlight the difference between positive and normative perspectives. Economists can test positive statements, but cannot test normative statements because they are value judgements.

- *..."if the government increases the sales tax on tobacco, people will purchase fewer cigarettes". This is a positive economic statement and can be tested. It is logically derived from a model called 'the law of demand', which states that people purchase less of a commodity when its price rises.*
- *..."the government should increase the tax on cigarettes". This is a normative statement made by a person who believes that higher cigarette taxes would be a good way to reduce tobacco consumption and improve the health of the population.*

It is the role of the economist to make positive statements about economic behaviour. Positive statements can be tested and used to build theories and models which can then be put into practice in developing policies from which everyone in the community can benefit. Economists can also express their opinion about economic issues and economic policy by making normative statements.

Scarcity, choice & opportunity cost

The economic problem is the problem of **relative scarcity** - resources are limited relative to society's unlimited wants. The economic problem applies everywhere, to everyone and for all time. It applies to individuals and to the community. It even applies to the wealthiest of individuals, such as Mark Zuckerberg, the cofounder of the social-networking website Facebook. Even though Mark is a billionaire, he still faces a time constraint and he still must make choices. The economic problem also applies to wealthy economies such as Australia. There are thousands of doctors in Australia, yet people are still forced to wait for elective surgery at public hospitals.

It is important not to confuse scarcity with poverty. Poverty can be reduced and even eliminated over time as a country's standard of living increases. But rising living standards cannot eliminate the problem of scarcity. The economic problem is ongoing and can never be ultimately solved. Once we satisfy one want, there will always be another to take its place.

The environment was once viewed as a **free good**, but as world population has increased, environmental goods have become more scarce. Many species of plants and animals are close to extinction as the pace of economic development has increased. In Australia, the demand for water has increased relative to supply. Most capital cities in Australia have imposed water restrictions in order to cope with the increasing scarcity of water. Part of the reason for the shortage of water is that the price of water has been kept very low. An economist would say that this gives the wrong signal or incentive to consumers - water is viewed as a relatively free good. If the price of water was increased in Australia to reflect its true scarcity value, then people would use it more carefully and water restrictions would no longer need to be imposed.

Simply stated, the economic problem means that there is not enough of everything for everybody! Relative scarcity means we are constantly forced to make choices between alternative uses - individuals decide how to allocate their income; societies must decide which goods and services will be produced and which resources will be used. The role of the economist is to study how and why people and groups make choices when confronted by relative scarcity. Making choices involves a trade-off - spending $20 to go to the movies means that you cannot spend it to do other things.

Every decision involves a choice between one course of action and another. In other words, every choice involves a trade-off, which means that every choice involves a cost. We often think of the costs of a good or service in dollar, or monetary terms, but this does not give a true indication of the true cost. The true or real cost of any decision should measure the value of the alternatives which have been foregone. **Opportunity cost** is the term used to represent the real or economic cost of a decision. It is best defined as the value of the best alternative that you give up. For example, the $1,200 used to purchase a new mobile phone could have been used to buy a laptop. Opportunity cost can be measured in monetary terms but it is important to think of it in terms of the actual opportunities that have been given up. By choosing the mobile phone, we have foregone the opportunity to purchase other goods, such as a laptop.

Opportunity cost

Every day, individuals make hundreds of choices - what to wear, how to get to school or work, what to buy for lunch, whether to study or go to training for sport. Make a list of five decisions you have made today that involved a trade-off. What decisions did your family make today that involved a tradeoff?

What federal or state government decisions were reported in the media? What was the tradeoff in each case?

There is no such thing as a 'free lunch!'

The expression "there is no such thing as a free lunch" reflects the relationship between scarcity and opportunity cost. If the government staged a free concert in the city, does that mean there is no cost? While the concert may be free for people to attend, from society's perspective, resources have been used to stage the concert. These scarce resources could have been used to provide other goods and services. Just because the concert has a 'zero' price does not make it a 'free good' - there is still an opportunity cost.

What is the opportunity cost of enrolling at university? It includes the monetary cost of fees and textbooks, but it also includes the value of your alternative use of time, such as getting a job. The value of time is an important part of opportunity cost. The opportunity cost of going to the movies includes more than just the price of the movie ticket! It includes the value of the two hours that you give up to watch the movie. If McDonalds announced that on a particular day they would give away free hamburgers, does this mean that the opportunity cost of consuming a hamburger would be zero? The answer is no - while the monetary cost is zero, you would have to sacrifice your valuable time waiting in the very long queue. Who would be more likely to stand in the queue - a student or a business executive? You would expect to see more students because the value of their time would be far less than someone earning a high salary.

Allocating scarce resources

What is the best way to allocate society's scarce resources? The economist compares the benefits of using resources against the cost. For example, the following table lists the benefits and costs associated with allocating resources to public hospitals. Each hospital costs $50 million to build, while the total benefits of increasing the number of hospitals increases but at a declining rate.

No. of Hospitals	Total Benefits $m	Total Costs $m	Net Benefits $m
1	100	50	50
2	180	100	80
3	240	150	90
4	280	200	80
5	300	250	50

For example, the total benefits of one hospital equals $100 million, the total benefits of two hospitals is $180 million, and for three it is $240 million. Why don't the benefits increase at a constant rate? Its simply because as you consume more of something the extra or additional benefit you get declines. It is a principle that applies to everything

that you consume. Its called the principle of decreasing marginal benefit. The fourth column in the table calculates the net benefits from each hospital - the difference between total benefits and total costs. Notice where the net benefits are largest. What is the correct or optimal number of hospitals? Should we keep allocating resources as long as total benefits exceed total costs? Should we build five hospitals? The answer is no. The correct answer is that we should keep allocating resources until we maximise net benefits. The economically efficient number of hospitals in this example is three. Because resources are scarce, then we should allocate them in such a way as to get the best value from their use. Net benefits are maximised at $90 million with three hospitals.

Economists like to use **marginal analysis** when making decisions about resource use. This mean calculating the marginal or additional benefit of each new hospital and compare it with the marginal cost (MC). Note that the marginal benefit of additional hospitals falls. The marginal benefit (MB) of the first hospital is $100 million, the MB of the second hospital is $80 million and the MB of the third hospital is just $60 million. The marginal cost (MC)of each hospital is constant at $50 million. It makes sense to allocate resources as long as the marginal benefits (MB) exceed the marginal costs (MC). In the example above, this occurs with three hospitals - MB equals $60 million and MC equals $50 million. It is inefficient to build the fourth hospital because the MB ($40 million) is less than the MC ($50 million). Comparing costs and benefits at the margin is a distinguishing feature of economic analysis. The cost benefit framework is a simple but very effective method to make decisions when allocating scarce resources.

No of Hospitals	Marginal Benefits $m	Marginal Costs $m
1	100	50
2	80	50
3	60	50
4	40	50
5	20	50

The production possibility frontier

An important economic model that economists use to illustrate the economic problem and the concept of opportunity cost is the **production possibility frontier** (PPF) model. The PPF shows all the combinations of goods and services that can be produced by an economy given the available resources and the level of technology. The model has some important assumptions:

- resources are fixed
- technology is fixed
- the economy produces just two goods - pizzas and cars

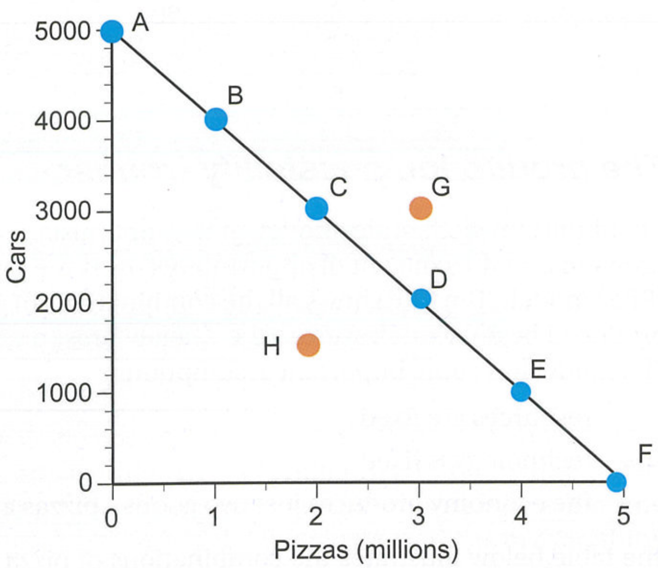
The table below illustrates the combinations of pizza and car production using the economy's scarce resources. If we convert this information into a graph then we obtain the production possibility frontier or curve. This is shown in Figure 1.3. Car production is measured on the vertical axis and pizza production is measured on the horizontal axis. The economy can produce anywhere along the frontier from point A to point F. At point A, all resources are being used to produce cars (5,000) and pizza production is zero. At point F, all resources are now being used to produce pizzas (5 million) and car production is zero.

Possibility	Pizzas (million)	Cars
A	0	5000
B	1	4000
C	2	3000
D	3	2000
E	4	1000
F	5	0

Notice the frontier illustrates the trade-off between producing pizzas and cars. To produce more pizzas, car production must decrease. This is because of scarcity - in other words, the frontier has a negative slope. The frontier also provides a measure of opportunity cost. The opportunity cost of producing each 1 million units of pizzas is 1 thousand cars. Alternatively, the opportunity cost of producing each 1,000 cars is 1 million pizzas. What is the opportunity cost of just 1 car? The answer is 1,000 pizzas. Alternatively we can determine the opportunity cost of producing just 1 pizza - it is 0.001 car. In this simple example, the opportunity cost along the frontier is constant and the PPF is a straight line. Can we produce at a point like G, outside the frontier? The answer is no, because we do not have enough resources. A point like H which is inside the frontier is attainable but it is inefficient. A point such as H would be associated with unemployed resources. It would be possible to increase the production of both goods, so point H is inferior compared to points on the frontier.

Figure 1.3 Simple production possibility frontier

The production frontier shows the maximum output for an economy given fixed resources and technology. The frontier illustrates scarcity, trade-offs and opportunity cost. The opportunity cost of 1 car is 1000 pizzas. This frontier is a straight line because opportunity cost is constant.



Will the PPF always be a straight line? Only if the opportunity cost between the two goods is constant. The normal shape for the PPF is to be 'bowed' outwards. This is because the opportunity cost of increasing the production of one good in an economy with scarce resources normally increases. It is referred to as the **law of increasing opportunity cost**. Constant opportunity cost means that resources are equally suited to producing all types of goods. In our example, it would mean that our resources are equally suited to producing both cars and pizzas. This would be highly unlikely. In reality, some resources would be better at making pizzas, while other resources would be more suited to manufacturing cars. When the productivity of resources differs between types of goods, then the PPF will be 'bowed' outwards. This is illustrated in the second example below.

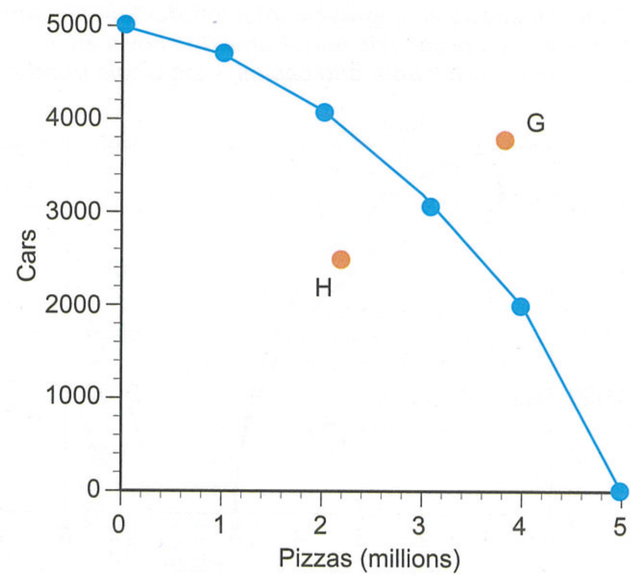
In this case the opportunity cost of the first 1 million pizzas is 400 cars (point A to B). But from point B to C the opportunity cost of the next 1 million pizzas increases to 600 cars, and from point C to D it increases to 1,000 cars. The opportunity cost of the fifth million pizzas is 1600 cars. This demonstrates the law of increasing opportunity cost - as production increases, the opportunity cost increases.

Possibility	Pizzas (millions)	Cars
A	0	5000
B	1	4600
C	2	4000
D	3	3000
E	4	1600
F	5	0

Figure 1.4 illustrates the typical 'bowed' out production frontier. Is there a best point to be on the production frontier? Any of the points A to F can be regarded as best, since each of these points is production efficient - resources are not being wasted. Ultimately, society must choose which combination of pizzas and cars it wants to consume. The particular combination of pizzas and cars that maximises net benefits

Figure 1.4 The law of increasing opportunity cost

This PPF is bowed outwards because of the law of increasing opportunity cost. As more pizzas are produced, the opportunity cost increases since resources are not equally productive at producing both cars and pizzas.



for society is known as the economically efficient point. Will society automatically choose this point? This is a very important question and we will be able to show that a free market economy will automatically produce at this point. Opportunity cost is involved in every single economic decision we make, so the production frontier is an important economic model. Imagine that the goods shown in figure 1.4 were not pizza and cars, but schools and hospitals, or consumer goods and capital goods. The principle still applies - to be able to build more schools, the government would have to forgo some hospitals. Every decision involves an opportunity cost because resources are scarce and there will always be a **tradeoff** - when a choice is made, something is foregone.

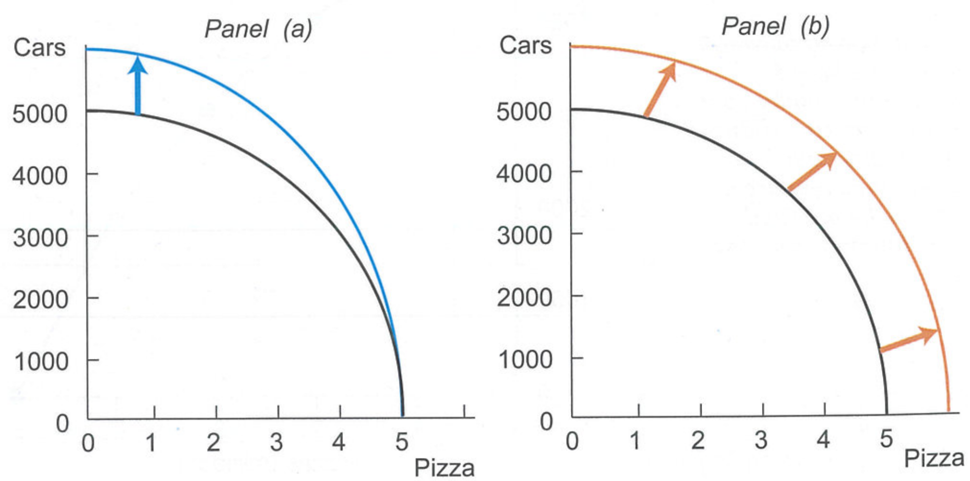
What happens to our model if we relax our initial assumptions regarding resources and technology? Figure 1.5 illustrates two possibilities. In panel a, the curve has shifted outwards along the car axis, perhaps as a result of an increase in the quantity of resources available for car production or the introduction of new technology in manufacturing cars. If instead there was an advance in pizza technology, then the frontier would move out along the horizontal axis. Panel b illustrates an outward shift of the whole frontier - more of both goods can be produced - there must have been an increase in the quantity or quality of resources that affects the production of both cars and pizza. Over time, an economy's production frontier will shift outwards as resources, such as the labour force and capital stock increase.

Economic growth

The production possibility frontier can be used to illustrate economic growth. Economic growth refers to an increase in the capacity of an economy to produce goods and services and is illustrated by panel b in figure 1.5. Economic growth can

Figure 1.5 Changes in the PPF

Panel (a) shows an improvement in production technology for cars - the frontier shifts along the car axis only. Panel (b) shows an increase in the quantity of resources (both labour and capital) - the whole frontier shifts out to the right.



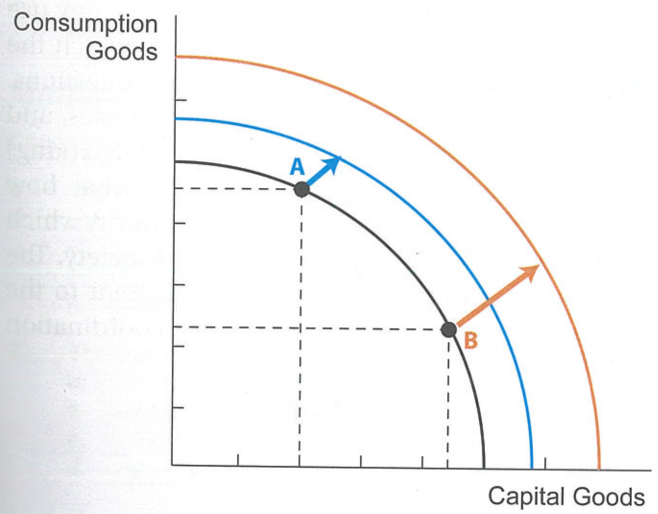
occur as a result of an increase in the quantity of resources or an improvement in their quality. When economies produce more goods and services it enables a higher standard of living which is the ultimate economic objective. But pursuing economic growth also involves a tradeoff - a tradeoff between present and future consumption. If we use most of our resources to produce consumer goods today, then that will mean potentially less consumption for the future. Sacrificing some current consumption enables resources to be used to produce capital goods.

Reducing current consumption increases savings which can be channeled into investment. Investment is seen as the 'engine' of economic growth. Investment is the creation of capital goods. Capital goods are man made goods that are used in the production process. They include machinery, factories, computers and transport networks. Higher economic growth enables higher future living standards but there is an opportunity cost - lower consumption today. Figure 1.6 helps to illustrate this tradeoff between present and future consumption. The PPF model below illustrates the choice between producing consumption goods or capital goods. If an economy chooses to locate at point A, where more resources are devoted to producing consumption goods, then the PPF only moves a short distance to the right. This reflects a lower rate of economic growth. If the economy instead chooses point B on the initial PPF, then, over time, the PPF moves much further to the right, indicating a faster rate of economic growth. In other words choices made today can have a significant impact on future living standards.

The economic system

Like individuals and groups, nations also face the economic problem. Collective wants such as education, health and public transport are unlimited, but resources are still relatively scarce. Thus a choice must be made as to which wants will be satisfied.

Figure 1.6 Capital goods and economic growth



If an economy chooses to produce at point A, its future economic growth will be less than for an economy that chooses point B. At point B, more resources are allocated to capital goods which promote economic growth. By choosing point B, the PPF moves much further to the right than for the economy located at point A.

1 Review

›nomics, when is a good scarce?

good and an economic good? Provide

:m of choice?

f resources.

cs?

ng, new capital goods are produced and new
'e resources said to be scarce?

and macroeconomics.

its purpose?